

Modeling Foreign Exchange Risk using GARCH-Type Models and Value-at-Risk

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Abstract

In this thesis we study whether GARCH-type models can give better estimates of foreign exchange market risk than a simple non-parametric Historical Simulation. We also want to understand which part of the model is responsible for the improvement: the time-varying volatility, the asymmetric reaction to shocks, or the fat-tailed and skewed distribution of the innovations. We use daily log-returns of three euro currency pairs (EUR/USD, EUR/GBP and EUR/CHF) from March 2015 to May 2026. For each pair we estimate nine models, which come from the combination of three variance models (GARCH, EGARCH, GJR-GARCH) with three innovation distributions (Gaussian, Student- t , skewed Student- t). We compute one-day-ahead Value-at-Risk (VaR) at the 95% and 99% levels out of sample, with a rolling window of 1,000 days (1,925 forecasts per model), and we evaluate the forecasts with the Kupiec, Christoffersen and Dynamic-Quantile tests. Our results show three main things. First, the innovation distribution is the most important choice: Student- t and skewed Student- t work much better than the Gaussian, both in-sample (AIC/BIC) and out-of-sample, while the Gaussian model does not cover the 99% tail well. Second, asymmetry does not help much. For EUR/USD the symmetric GARCH and GJR pass all the tests, but the asymmetric EGARCH is rejected, which is consistent with the fact that FX returns do not really have a leverage effect. Third, modelling the conditional volatility does help compared to Historical Simulation, which often fails the Dynamic-Quantile test. In general, a simple GARCH(1,1) or GJR-GARCH(1,1) with Student- t innovations gives the most reliable VaR. This is in line with Hansen and Lunde [7]: for exchange rates it is hard to beat a GARCH(1,1), and the important thing is the tail of the distribution and not the variance equation.

1 Introduction

Exchange rates can move fast and in an unexpected way, and this makes the measurement of their risk difficult. Daily returns of financial assets usually show two well-known properties. The first one is *volatility clustering*, which means that calm periods and turbulent periods tend to group together in time. The second one is *heavy tails*, which means that large losses happen more often than a normal distribution would predict. Value-at-Risk (VaR) is one of the most common measures of market risk. It gives the loss that is not exceeded over a given horizon with a given probability [9]. The problem is that the quality of a VaR estimate depends a lot on how we model the volatility and the tail of the distribution. If the model ignores volatility clustering, the VaR will be too low exactly when the risk is high. If the model assumes normality, it will underestimate how often big losses happen.

Research question. The main research question of this thesis is the following: can GARCH-type models improve the estimation of foreign exchange market risk compared to a simple Historical VaR approach, and how much does the choice of the conditional-variance specification and of the innovation distribution influence the accuracy of the VaR forecasts? We study this question in two directions. In the first direction we compare a non-parametric Historical Simulation with the parametric GARCH-based forecasts. This tells us if modelling the conditional

volatility is useful at all. In the second direction we stay inside the parametric family and we try to separate two effects: the effect of the asymmetric volatility (EGARCH and GJR-GARCH against the symmetric GARCH) and the effect of the fat-tailed and skewed innovations (Student- t and skewed Student- t against the Gaussian). In this thesis the word “improve” means a better out-of-sample backtesting performance (correct coverage, no clustering of the violations and no predictability of the violations), and not a better in-sample fit.

Relation to the literature. Our work is based on the ARCH/GARCH literature. The GARCH model was introduced by Bollerslev [1]. The asymmetric extensions were proposed by Nelson [11] for the EGARCH and by Glosten et al. [5] for the GJR-GARCH. The idea of using fat-tailed innovations goes back to Bollerslev [2], and the skewed Student- t distribution that we use is the one of Hansen [6]. For the evaluation we use the coverage tests of Kupiec [10] and Christoffersen [3] and the Dynamic-Quantile test of Engle and Manganelli [4]. The most important reference for us is Hansen and Lunde [7]. They compared a very large number of volatility models and they found that the leverage effect is important for equity indices, but for exchange rates nothing really beats a simple GARCH(1,1). We do not propose a new model. Our contribution is an out-of-sample VaR comparison (a “horse-race”) with a full factorial design on a recent euro sample (2015–2026), a period that includes the Brexit vote in 2016, the COVID shock in 2020 and the UK gilt crisis in 2022. With this design we can separate the roles of the variance dynamics, the asymmetry and the shape of the tail.

Main results. Our main results can be summarized in the following way. Modelling the conditional volatility helps: Historical Simulation gives more or less the correct number of violations, but it often fails the Dynamic-Quantile test, which means that its violations are still predictable. Inside the parametric family, the innovation distribution is the most important element. Student- t and skewed Student- t are preferred by AIC and BIC for all three pairs, and they give a reliable 99% VaR, while the Gaussian versions do not cover the deep tail well. The asymmetry, on the other side, does not bring much. For EUR/USD the symmetric GARCH and GJR pass all the tests at both confidence levels, but the EGARCH is rejected, and the estimated leverage parameters are very small. The model that is most robust across the pairs and the confidence levels is a GARCH(1,1) or a GJR-GARCH(1,1) with Student- t (or skewed Student- t) innovations.

The rest of the thesis is organized as follows. Section 2 describes the models, the innovation distributions, the VaR computation and the backtests. Section 3 presents the data and the descriptive analysis. Section 4 reports the estimation, the goodness of fit and the backtesting results. Section 5 concludes and discusses the limitations and some possible extensions.

2 Methodology and Empirical Strategy

2.1 Returns and the conditional-mean equation

Let P_t be the daily closing exchange rate. We work with percentage log-returns

$$r_t = 100 \ln\left(\frac{P_t}{P_{t-1}}\right). \quad (1)$$

Log-returns are (approximately) stationary, they are additive over time and they are comparable across the three pairs. The factor 100 is only a scaling that makes the numerical optimization of the likelihood more stable. Because daily FX returns have almost no predictable drift (see Section 3), we use a constant conditional mean:

$$r_t = \mu + u_t, \quad u_t = \sigma_t z_t, \quad z_t \sim D(0, 1; \theta), \quad (2)$$

where σ_t^2 is the conditional variance, z_t is an i.i.d. standardized innovation with unit variance that follows a distribution D with shape parameters θ , and u_t is the residual of the mean equation.

2.2 Conditional-variance specifications

We estimate three specifications for the variance. The first one is the symmetric **GARCH(1,1)** of Bollerslev [1],

$$\sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (3)$$

where α is the ARCH (news) coefficient, β is the GARCH (persistence) coefficient, and the sum $\alpha + \beta$ measures how persistent the volatility is. The second one is the **EGARCH(1,1)** of Nelson [11],

$$\ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2, \quad (4)$$

which models the log of the variance, so the variance stays positive without putting constraints on the parameters. Here the term γz_{t-1} captures the asymmetry: if γ is negative, then bad news increase the volatility more than good news of the same size (this is the leverage effect). The third one is the **GJR-GARCH(1,1)** of Glosten et al. [5],

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{I}[u_{t-1} < 0])u_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (5)$$

which adds an extra reaction of the variance after a negative shock, and the leverage effect is present when $\gamma > 0$. In this way the EGARCH and the GJR-GARCH both contain the symmetric case, so we can test if the leverage effect that is known from equity markets also exists in FX.

2.3 Innovation distributions

Each variance model is combined with three distributions for z_t . They add, one by one, the stylized facts of the returns (see Table 1):

1. **Gaussian**, $z_t \sim N(0,1)$. This is the benchmark. It does not capture heavy tails or skewness.
2. **Student- t** with $\nu > 2$ degrees of freedom (standardized to unit variance). It captures symmetric heavy tails; a smaller ν means fatter tails.
3. **Skewed Student- t** [6], with degrees of freedom η and a skewness parameter λ . It captures heavy tails and skewness of the innovations at the same time.

The three variance models times the three distributions give a 3×3 design of nine parametric specifications per currency pair (27 model fits in total). We estimate all of them by (quasi-)maximum likelihood, using the `arch` package in Python.

Table 1: The nine model–distribution specifications and the stylized features that each one can capture.

Variance model	Innovation distribution	Asymmetry	Heavy tails	Skewness
GARCH(1,1)	Gaussian	No	No	No
GARCH(1,1)	Student- t	No	Yes	No
GARCH(1,1)	Skewed Student- t	No	Yes	Yes
EGARCH(1,1)	Gaussian	Yes	No	No
EGARCH(1,1)	Student- t	Yes	Yes	No
EGARCH(1,1)	Skewed Student- t	Yes	Yes	Yes
GJR-GARCH(1,1)	Gaussian	Yes	No	No
GJR-GARCH(1,1)	Student- t	Yes	Yes	No
GJR-GARCH(1,1)	Skewed Student- t	Yes	Yes	Yes

2.4 Value-at-Risk

For a long position, the one-day-ahead VaR at the coverage level α is the conditional α -quantile of the return distribution:

$$\text{VaR}_{\alpha,t} = \mu + \sigma_t q_\alpha(\theta), \quad (6)$$

where $q_\alpha(\theta)$ is the (negative) α -quantile of the standardized innovation distribution (for example the $N(0,1)$ quantile in the Gaussian case, or the standardized Student- t quantile in the other cases). We report $\alpha \in \{5\%, 1\%\}$, that is the 95% and the 99% VaR. Equation (6) shows clearly that VaR has two ingredients: the conditional volatility σ_t (which comes from the variance model) and the tail quantile q_α (which comes from the innovation distribution). These are exactly the two dimensions that we want to compare. As a non-parametric benchmark we also compute **Historical Simulation** VaR, which is simply the empirical α -quantile of the returns inside the rolling window and does not assume any distribution or any volatility dynamics.

2.5 Out-of-sample design

All the VaR forecasts are computed out of sample. We use a fixed-size *rolling window* of $W = 1,000$ trading days. At every step the models are estimated on the last W observations and used to forecast the VaR of the next day. To make the computation feasible, we re-estimate the parameters every 5 steps and we keep them fixed in between. This gives 1,925 one-day-ahead forecasts per model and per pair, which cover the period from January 2019 to May 2026.

2.6 Backtesting

We define the violation indicator as

$$I_t = \mathbb{I}(r_t < \text{VaR}_{\alpha,t}), \quad (7)$$

which is equal to one when the realized return is below the forecasted VaR. If the model is correct, the sequence $\{I_t\}$ should be i.i.d. Bernoulli with probability α . We use three tests, and each test is more demanding than the previous one.

- **Kupiec** unconditional coverage [10]: a likelihood-ratio test that checks if the observed violation rate is equal to α . $LR_{uc} \sim \chi^2(1)$. It looks only at the *number* of violations.
- **Christoffersen** conditional coverage [3]: it adds an independence test against first-order clustering, $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$. It checks the number of violations *and* that the violations do not come one after another.
- **Dynamic Quantile (DQ)** [4]: a Wald test in a regression of the demeaned hit series on its own lags. It checks that the violations are not predictable from the past information. This is the strictest test.

In all three tests the null hypothesis is that “the model is correctly specified”. So a *pass* is a high p -value (we use the 5% level: we do not reject when $p \geq 0.05$). To pass all three tests at the same time is a strong requirement. A model that fails the DQ test (as Historical Simulation usually does) produces violations that a risk manager could have predicted in advance.

3 Data and Descriptive Analysis

Sample. We use daily closing quotations from Yahoo Finance for the three euro pairs EUR/USD, EUR/GBP and EUR/CHF, from 2 March 2015 to 29 May 2026. We downloaded the data with

the `yfinance` library. After the alignment of the dates we have 2,928 prices and 2,925 daily log-returns per pair. Using the euro as the common leg lets us isolate the other currency, and the three pairs cover different levels of liquidity and risk: EUR/USD is the most liquid and almost symmetric pair, EUR/GBP is event-driven and has the sharpest jumps, and EUR/CHF is the third one. We start the sample after the Swiss National Bank removed the 1.20 EUR/CHF floor on 15 January 2015. By starting in March 2015 we remove two problems at the same time: the very low volatility of the managed peg before 2015, and the single de-pegging jump of about -19% . In this way EUR/CHF enters the study as a clean floating series and not as a structural break. The horizon is one trading day, which is also the horizon used in the regulation for VaR.

Return behaviour. Figure 1 shows the three return series. All of them have the typical volatility-clustering pattern, with quiet periods and then bursts of activity. This is clearly visible around the 2020 COVID shock (EUR/USD) and around the 2022 gilt crisis (EUR/GBP, where the two-day swing of about $\pm 11\%$ in October 2022 is much larger than everything else in the sample). Figure 2 shows the squared returns, which are a proxy for the realized variance. We can see that the large squared returns are not isolated but come close to each other in time, and this is a direct visual sign of ARCH effects, so a conditional-variance model is justified. Figure 3 shows only the top-1% absolute returns of EUR/GBP, and it makes clear that the risk of this pair is concentrated in a few extreme days.

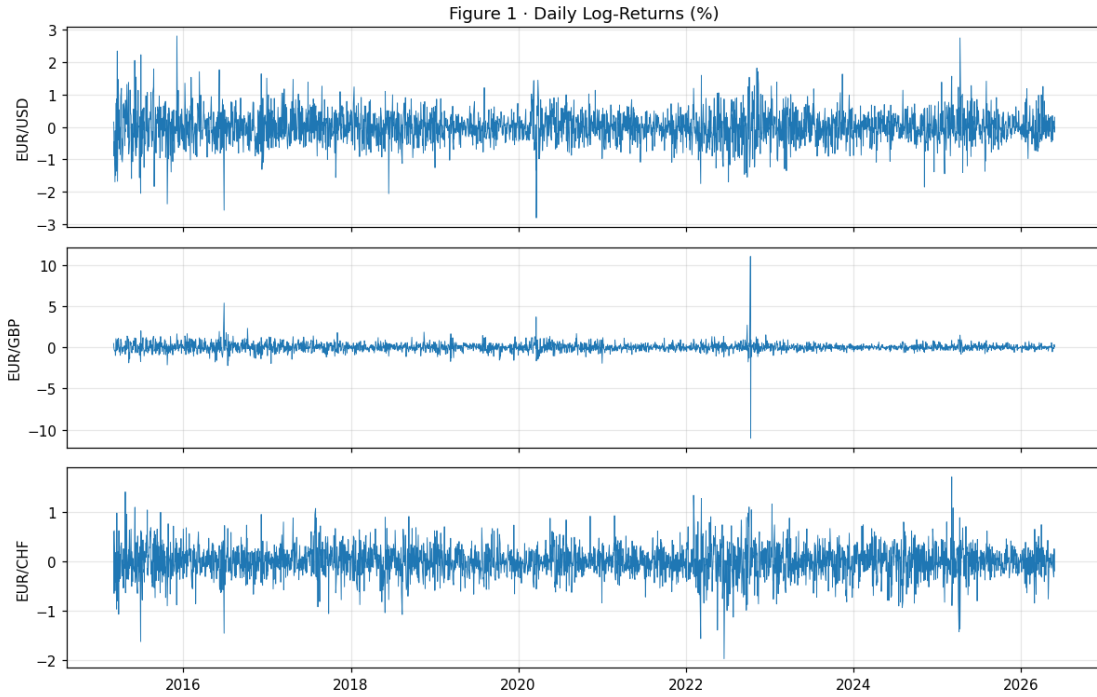


Figure 1: Daily log-returns (%) of EUR/USD, EUR/GBP and EUR/CHF, March 2015–May 2026. All the series show volatility clustering; EUR/GBP has the largest isolated jumps (October 2022).

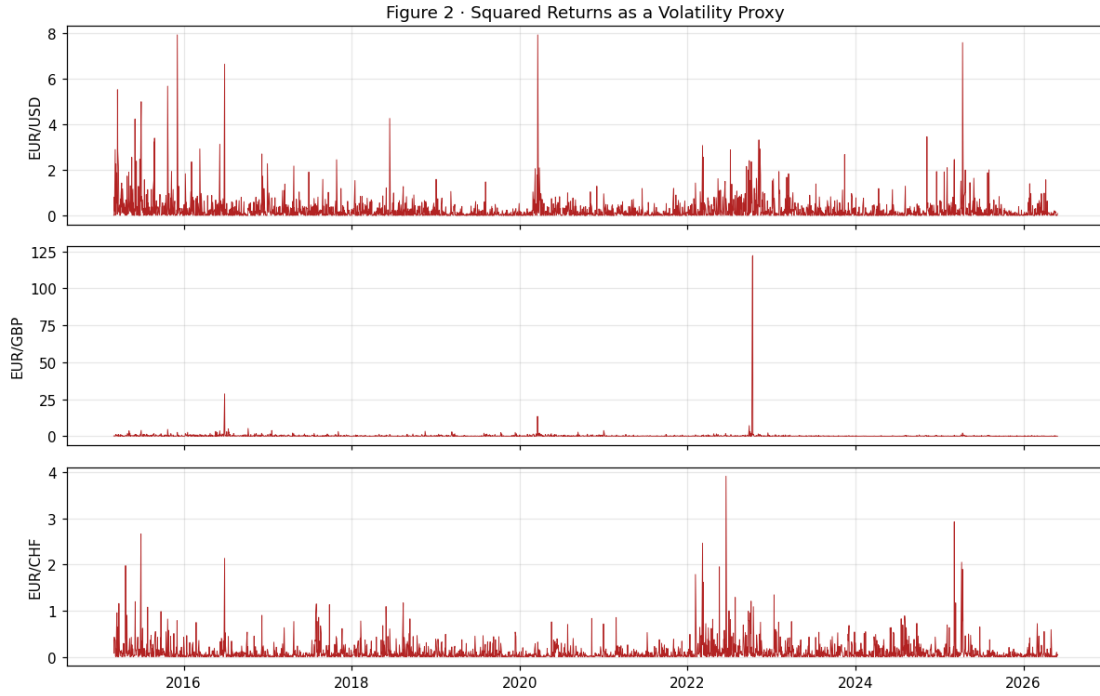


Figure 2: Squared returns as a proxy for the volatility. The fact that the large squared returns cluster in time is a direct sign of autoregressive conditional heteroskedasticity (ARCH effects).

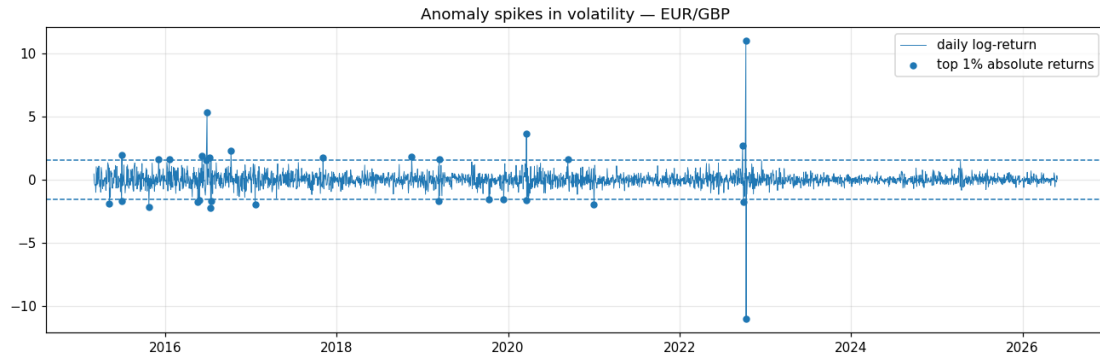


Figure 3: Anomaly spikes of EUR/GBP: the daily log-returns with the top-1% absolute returns marked with dots and the ± 99 th-percentile thresholds (dashed lines). The tail risk is concentrated in a few days, especially the October-2022 gilt-crisis swing, which is the reason why EUR/GBP is the most difficult pair to model.

Descriptive statistics. Table 2 reports the descriptive statistics of the returns. There are three points that we want to highlight. First, all the means are very close to zero (only some thousandths of a percent per day), which justifies the constant-mean equation (2). Second, all the series are strongly non-normal: the Jarque–Bera test rejects normality at any usual level ($p < 10^{-4}$). This is caused by the excess kurtosis, which is moderate for EUR/USD (2.67) and EUR/CHF (2.40), but very large for EUR/GBP (114.7), where the October-2022 outliers dominate the fourth moment. Third, the skewness is different for each pair and mostly small (EUR/GBP is right-skewed with 0.45, the other two are almost symmetric). This already suggests that the *skewness* will be less important than the *heavy tails*. It is also interesting that, once we remove the SNB break, EUR/CHF becomes the *calmest* pair, with the lowest standard deviation (0.32). This is the opposite of its usual reputation as a tail-risk currency. Finally, the augmented Dickey–Fuller test rejects the unit root for all three series (Table 2), so the log-returns

are stationary and can be used in a GARCH model.

Table 2: Descriptive statistics of the daily log-returns (%), March 2015–May 2026. Kurtosis is the excess kurtosis (normal = 0). JB is the Jarque–Bera normality test; ADF is the augmented Dickey–Fuller unit-root test (constant, lag length chosen by AIC). *** means rejection of the corresponding null at 1%.

Pair	Mean	Std	Min	Max	Skew	Ex. kurt.	JB p	ADF stat	ADF p
EUR/USD	0.0015	0.489	−2.81	2.81	0.09	2.67	0.000***	−23.95***	0.000
EUR/GBP	0.0060	0.550	−11.05	11.01	0.45	114.69	0.000***	−29.76***	0.000
EUR/CHF	−0.0052	0.319	−1.98	1.71	−0.13	2.40	0.000***	−20.76***	0.000

Evidence for fat-tailed innovations. One question is whether the heavy tails of Table 2 come only from the volatility clustering (which a GARCH model already removes) or if they are a real property of the *innovations* z_t . Figure 4 answers this question with quantile–quantile plots of the standardized residuals of a Gaussian GARCH(1,1) fit. Even after we condition on the estimated volatility, the residuals are still far from the $N(0, 1)$ line and show the typical fat-tailed “S” shape (left column), while a standardized Student- t with a low degrees of freedom (around $\nu = 5$) fits them well (center-right columns). So the excess kurtosis is really in the innovations and not only in the volatility dynamics. This is exactly what a fat-tailed distribution is made for, and it is the reason why we use the Student- t and the skewed Student- t .

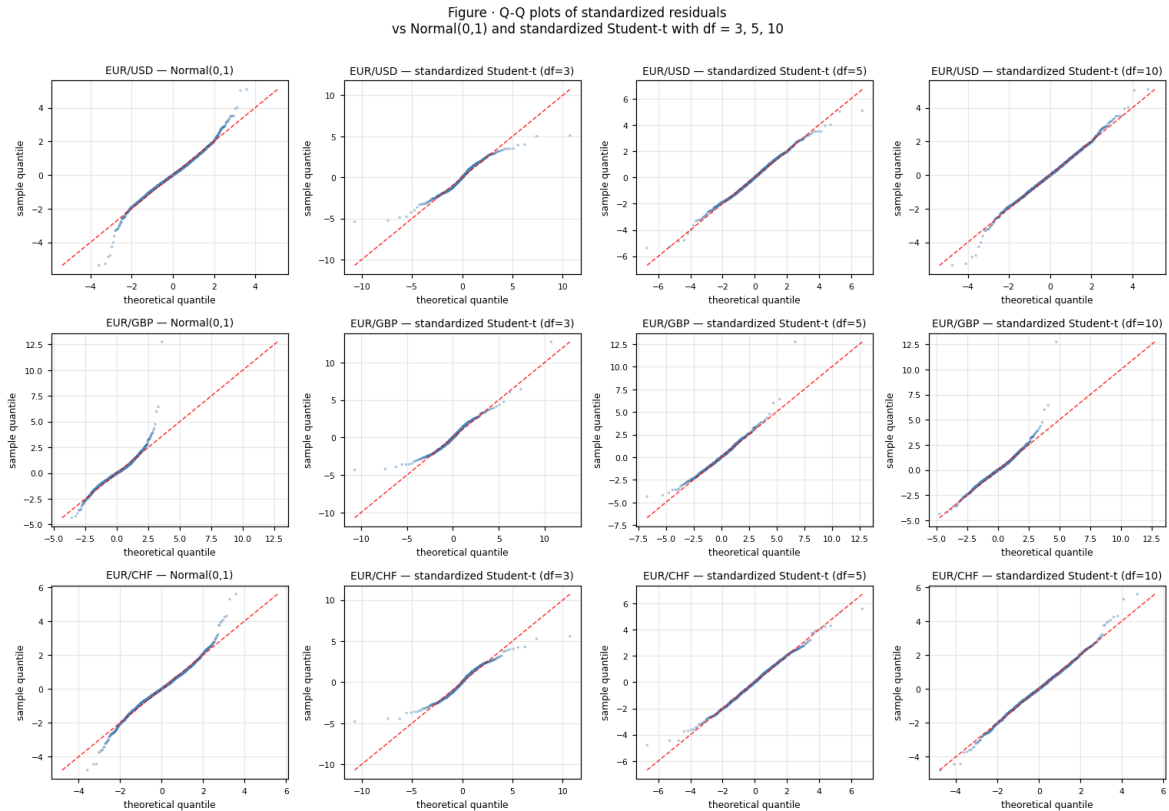


Figure 4: Q–Q plots of the standardized GARCH(1,1) residuals against the $N(0,1)$ distribution and against the standardized Student- t with $\nu = 3, 5, 10$, for each pair (rows). After the volatility clustering is removed, the residuals are still heavy-tailed compared to the normal (first column) but they fit well a Student- t around $\nu = 5$. This justifies the use of fat-tailed innovations.

4 Empirical Results

4.1 Estimated conditional volatility

Figure 5 shows the estimated annualized conditional volatility of the three variance models (in the Student- t version) for each pair. Two things can be observed. First, the three models give almost the same path. The GARCH, EGARCH and GJR lines are practically on top of each other, except for very short spikes. This already suggests that the choice *between* the variance dynamics is not so important. Second, the estimated volatility reacts in a reasonable way to the events that we know: the 2020 COVID spike in EUR/USD and the very large but short spike in EUR/GBP in autumn 2022. Different from earlier studies on the pre-2015 peg, here the EGARCH stays numerically stable during all the sample, because the extreme SNB de-pegging jump, which was destabilizing its exponential recursion before, is now outside of the sample.

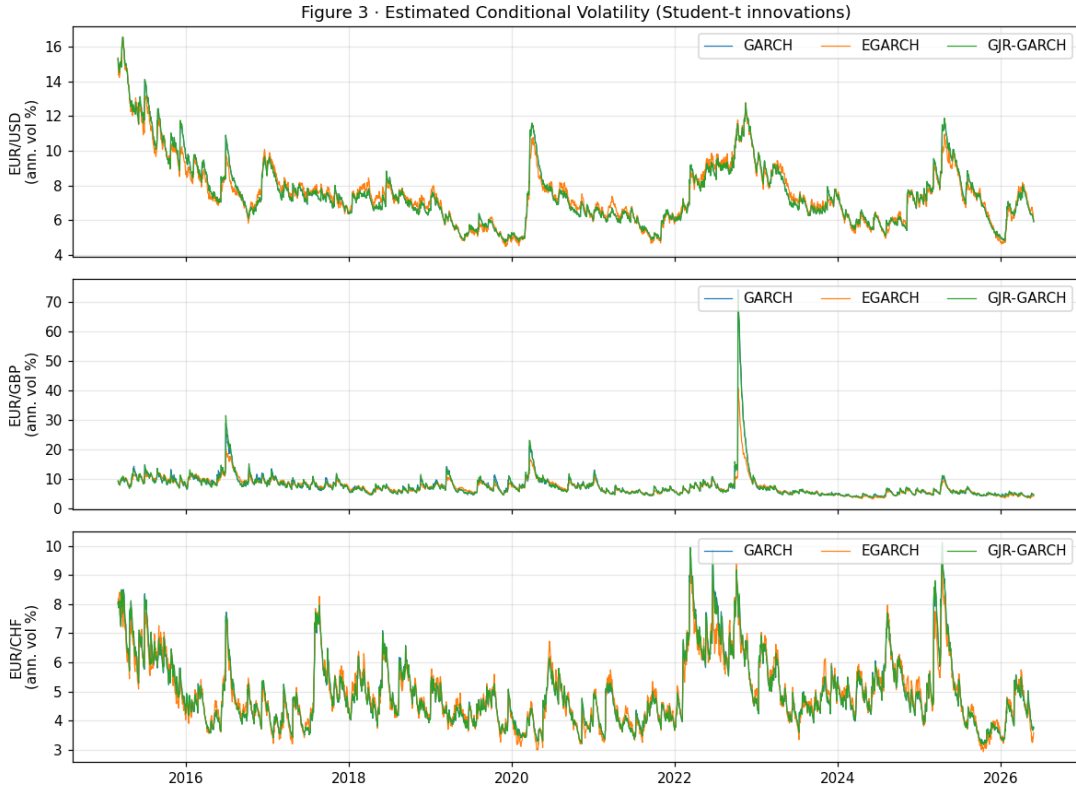


Figure 5: Estimated annualized conditional volatility (%) from GARCH, EGARCH and GJR-GARCH with Student- t innovations. The three specifications are almost identical, so the variance *dynamics* are similar between the models. The differences in the VaR will therefore come mainly from the tail of the innovation distribution.

4.2 Parameter estimates

Table 3 reports, for each pair, the specification chosen by AIC and its main parameters. We can see some regular patterns. The volatility is very persistent everywhere: $\alpha + \beta \approx 0.99$ for EUR/USD (GARCH) and 0.98 for EUR/CHF, and the log-variance persistence of the EGARCH for EUR/GBP is $\beta = 0.986$. So the shocks disappear slowly and the turbulence stays for several weeks. The *tail* parameters are clearly in the heavy-tail region: $\nu \approx 7.2$ for EUR/USD, 6.5 for EUR/CHF and only 4.9 for EUR/GBP. This confirms that the innovations are fat-tailed and not Gaussian, and that EUR/GBP has the fattest tails. The *asymmetry* parameters, on the contrary, are very small: the EGARCH sign term is $\gamma \approx -0.005$ for EUR/USD and $+0.038$ for

EUR/GBP, and the GJR leverage terms (not reported here) are also close to zero and do not have a consistent sign across the pairs. So FX volatility is persistent and fat-tailed, but it does not have much of the leverage effect that we know from equity markets. This is the same pattern that Hansen and Lunde [7] describe.

Table 3: Estimated parameters of the specification chosen by AIC for each pair (full-sample fit). α is the ARCH/magnitude coefficient, β is the persistence coefficient, γ is the asymmetry coefficient, ν is the Student- t degrees of freedom, and λ is the skewness parameter. A dash means that the parameter is not in the specification.

Pair	Best model (AIC)	ω	α	β	γ	ν	λ	$\alpha + \beta$
EUR/USD	GARCH / Student- t	0.0016	0.031	0.962	—	7.17	—	0.992
EUR/GBP	EGARCH / Skewed- t	−0.017	0.123	0.986	0.038	4.94	0.082	—
EUR/CHF	GARCH / Student- t	0.0019	0.057	0.925	—	6.46	—	0.982

4.3 Goodness of fit

Table 4 ranks the specifications by their in-sample fit. For all three pairs the best model uses a *fat-tailed* distribution. The improvement when we replace the Gaussian by a Student- t is large, about 130 AIC points for both EUR/USD and EUR/CHF, which is much more than the cost of two points for the extra parameter. The gain from the *asymmetry* is smaller: for EUR/USD and EUR/CHF the symmetric GARCH/Student- t is better than its EGARCH and GJR versions, and only for EUR/GBP (the pair with the most extreme jumps and the fattest tails) the asymmetric EGARCH/Skewed- t wins. The BIC, which penalizes the number of parameters more, gives the same ranking as the AIC in all the cases. But we have to keep in mind that the in-sample fit is not our final goal. The information criteria measure the description of the past data, not the out-of-sample accuracy of the risk forecast, which is what a VaR model has to give. The decision is made by the backtests.

Table 4: Goodness of fit of the specification chosen by AIC for each pair (log-likelihood, AIC, BIC). In all the cases a Student- t -type distribution minimizes the information criteria; the fat-tailed distribution improves the AIC with respect to the Gaussian by about 130 points.

Pair	Best model (AIC)	Log-likelihood	AIC	BIC
EUR/USD	GARCH / Student- t	−1839.62	3689.25	3719.15
EUR/GBP	EGARCH / Skewed- t	−1523.98	3061.96	3103.83
EUR/CHF	GARCH / Student- t	−606.29	1222.58	1252.48

4.4 Value-at-Risk forecasts

Figure 6 shows the 95% and 99% VaR forecasts of the GARCH/Student- t model together with the realized returns for each pair, and the 99% violations are marked. We can see that the VaR bands become wider in the turbulent periods and narrower in the calm periods, which is exactly the advantage of the conditional volatility. The violations are few and spread in time, not grouped, and this is confirmed later by the formal backtests. The EUR/GBP panel shows well the main problem of the deep tail: the single spike of October 2022 is so large that no smooth volatility band can contain it. So for GBP the adequacy of the tail depends on the innovation distribution and not on the volatility path.

Figure 5. Value-at-Risk forecasts at the 95% and 99% confidence levels

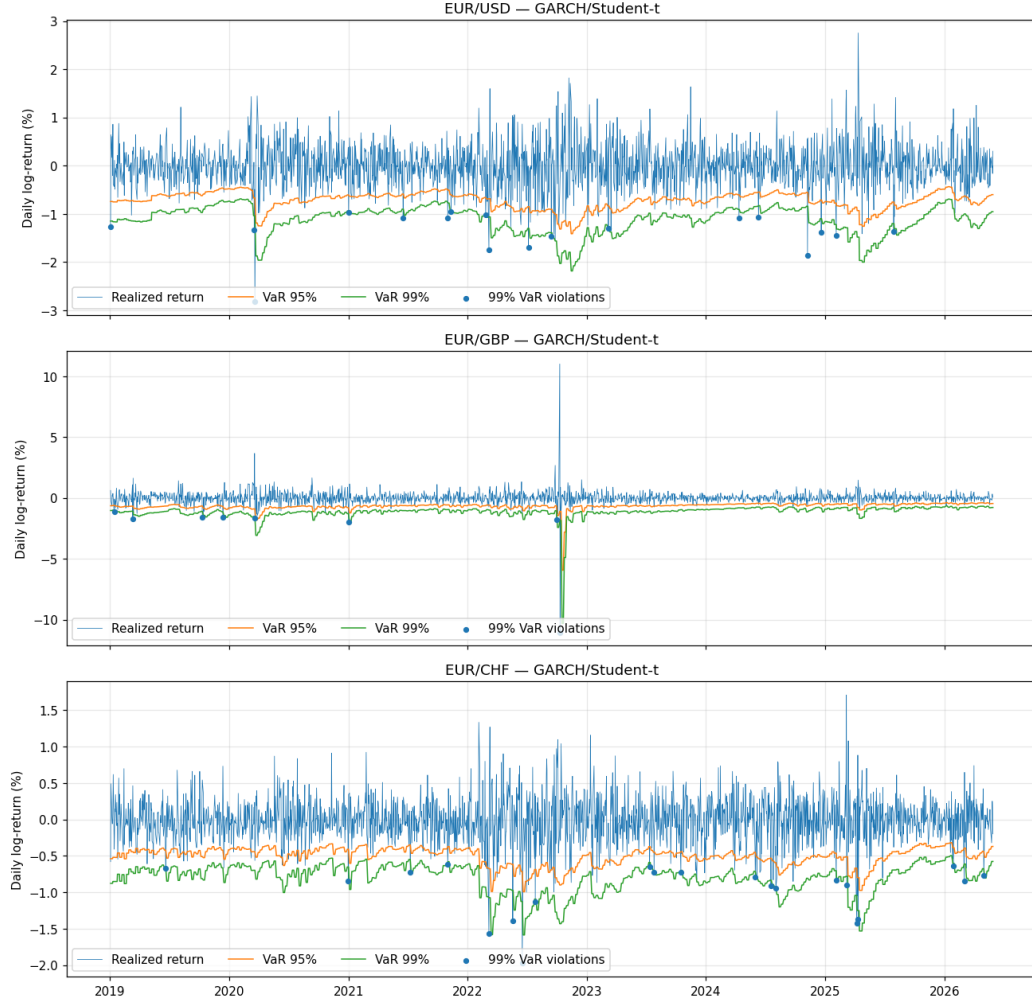


Figure 6: Out-of-sample 95% and 99% VaR forecasts of GARCH/Student- t (orange/green) against the realized returns (blue), 2019–2026; the blue dots are the 99% violations. The bands follow the time-varying volatility and the violations are spread and not clustered.

4.5 Backtesting results

Tables 5–7 report the out-of-sample backtests for all the ten specifications (nine parametric plus Historical Simulation) at both confidence levels. The p -values in bold are the passes (not rejected at 5%), and the last column of each block counts how many of the three tests are passed. We discuss the three pairs one by one.

EUR/USD (Table 5). This liquid and almost symmetric pair is the clearest case. The symmetric **GARCH** and **GJR** models with Student- t or skewed Student- t innovations pass all three tests at *both* the 95% and the 99% level. The Gaussian versions are fine at 95% but they fail the DQ test at 99%. And, what is more surprising, all three **EGARCH** specifications are rejected: they have too many violations in the deep tail (for example EGARCH/Gaussian has a violation rate of 2.3% against a target of 1% at the 99% level). So the EGARCH, which is competitive by AIC, is the *worst* model out of sample for EUR/USD. This is a good example of the fact that the model with the best fit is not always the best risk forecaster. Historical Simulation passes at 95% but, like the Gaussian models, it fails the DQ test at 99%.

EUR/GBP (Table 6). The event-driven pair is the most difficult one, and here the difficulty is at the 95% level: most of the models are too conservative. They overestimate the risk in the normal periods, so the number of violations is too small and Kupiec rejects them. Only the AIC-preferred **EGARCH/Skewed- t** passes all three tests at 95%. At the 99% level the situation is different, and five specifications pass all three tests, because at this deeper quantile the daily VaR band is wide enough to contain the extreme October-2022 loss. Historical Simulation fails again (Kupiec and DQ at both levels), because its static empirical quantile is too rough for the jump-prone dynamics of GBP. GBP is the only pair where the *asymmetric* model is useful, but the reason is the skewed- t tail and not the EGARCH sign term.

EUR/CHF (Table 7). Now that it is a calm and well-behaved series, EUR/CHF is easy, and several specifications pass. Again the important difference is the distribution: the **Gaussian** GARCH and GJR are rejected by Kupiec at 99% (too many tail violations), while the **Student- t** and **skewed Student- t** versions of every variance model pass at 99%, and GARCH/Student- t , GARCH/Skewed- t and GJR/Skewed- t pass at both levels. Historical Simulation fails again the DQ test at both levels, which shows that its violations are still predictable even for a quiet pair.

Summary. Three conclusions are common to all the pairs. (i) Modelling the conditional volatility helps: Historical Simulation has more or less the correct count but it fails the dynamic-adequacy (DQ) test in five of the six pair-level combinations. (ii) The innovation distribution is the most important parametric choice: the Gaussian models fail the 99% tail, while the Student- t and skewed Student- t models pass it for all the pairs. (iii) The asymmetry adds little and can even make things worse: the symmetric GARCH/GJR with fat tails is the most reliable combination, the EGARCH is not reliable for EUR/USD, and the asymmetry helps only for the jump-prone EUR/GBP (and there through the skewed tail). If a risk manager has to choose one model for these euro pairs, a simple GARCH(1,1) (or GJR-GARCH(1,1)) with Student- t innovations is a good choice.

Table 5: VaR backtesting, **EUR/USD**. Violation rate and p -values of the Kupiec, Christoffersen (Christof.) and Dynamic-Quantile (DQ) tests at the 95% and 99% levels (1,925 forecasts; target rates 5% and 1%). Numbers in **bold** are not rejected at the 5% level (the test is passed); the column “#” counts how many of the three tests are passed (0–3).

Specification	95% VaR					99% VaR				
	Rate	Kupiec	Christof.	DQ	#	Rate	Kupiec	Christof.	DQ	#
GARCH / Gaussian	0.0447	0.275	0.316	0.410	3	0.0135	0.142	0.242	0.011	2
GARCH / Student- t	0.0473	0.580	0.383	0.502	3	0.0094	0.772	0.817	0.409	3
GARCH / Skewed- t	0.0473	0.580	0.383	0.502	3	0.0088	0.599	0.755	0.333	3
EGARCH / Gaussian	0.0525	0.622	0.252	0.001	2	0.0229	0.000	0.000	0.000	0
EGARCH / Student- t	0.0571	0.159	0.033	0.000	1	0.0182	0.001	0.000	0.000	0
EGARCH / Skewed- t	0.0608	0.036	0.000	0.000	0	0.0208	0.000	0.000	0.000	0
GJR / Gaussian	0.0462	0.443	0.370	0.298	3	0.0130	0.208	0.330	0.011	2
GJR / Student- t	0.0488	0.813	0.366	0.485	3	0.0094	0.772	0.817	0.409	3
GJR / Skewed- t	0.0488	0.813	0.366	0.485	3	0.0099	0.954	0.834	0.476	3
Historical VaR	0.0483	0.733	0.718	0.622	3	0.0130	0.208	0.326	0.010	2

Table 6: VaR backtesting, **EUR/GBP**. See Table 5 for the layout. At 95% most of the models are too conservative (Kupiec rejects them); only EGARCH/Skewed- t passes all three. At 99% five specifications pass; Historical Simulation fails at both levels.

Specification	95% VaR					99% VaR				
	Rate	Kupiec	Christof.	DQ	#	Rate	Kupiec	Christof.	DQ	#
GARCH / Gaussian	0.0332	0.000	0.001	0.020	0	0.0094	0.772	0.809	0.981	3
GARCH / Student- t	0.0343	0.001	0.003	0.019	0	0.0042	0.004	0.014	0.247	1
GARCH / Skewed- t	0.0384	0.016	0.053	0.189	2	0.0062	0.074	0.189	0.711	3
EGARCH / Gaussian	0.0410	0.063	0.132	0.001	2	0.0109	0.693	0.445	0.253	3
EGARCH / Student- t	0.0353	0.002	0.007	0.014	0	0.0047	0.009	0.031	0.351	1
EGARCH / Skewed- t	0.0431	0.156	0.347	0.133	3	0.0062	0.074	0.036	0.069	2
GJR / Gaussian	0.0348	0.001	0.005	0.006	0	0.0099	0.954	0.826	0.978	3
GJR / Student- t	0.0332	0.000	0.001	0.009	0	0.0042	0.004	0.014	0.247	1
GJR / Skewed- t	0.0369	0.006	0.021	0.093	1	0.0068	0.128	0.288	0.810	3
Historical VaR	0.0312	0.000	0.000	0.000	0	0.0047	0.009	0.031	0.007	0

Table 7: VaR backtesting, **EUR/CHF**. See Table 5 for the layout. The Gaussian variance models fail Kupiec at 99%; the Student- t and skewed Student- t versions of every model pass at 99%. Historical Simulation fails the DQ test at both levels.

Specification	95% VaR					99% VaR				
	Rate	Kupiec	Christof.	DQ	#	Rate	Kupiec	Christof.	DQ	#
GARCH / Gaussian	0.0494	0.896	0.160	0.164	3	0.0161	0.013	0.028	0.058	1
GARCH / Student- t	0.0494	0.896	0.554	0.391	3	0.0109	0.693	0.734	0.581	3
GARCH / Skewed- t	0.0530	0.551	0.441	0.388	3	0.0104	0.864	0.799	0.570	3
EGARCH / Gaussian	0.0478	0.654	0.105	0.071	3	0.0156	0.023	0.047	0.000	0
EGARCH / Student- t	0.0499	0.979	0.179	0.049	2	0.0104	0.864	0.799	0.570	3
EGARCH / Skewed- t	0.0525	0.622	0.057	0.010	2	0.0104	0.864	0.799	0.570	3
GJR / Gaussian	0.0514	0.775	0.110	0.087	3	0.0161	0.013	0.028	0.058	1
GJR / Student- t	0.0514	0.775	0.110	0.123	3	0.0104	0.864	0.799	0.570	3
GJR / Skewed- t	0.0535	0.485	0.139	0.087	3	0.0114	0.538	0.641	0.571	3
Historical VaR	0.0545	0.367	0.001	0.000	1	0.0125	0.295	0.427	0.000	2

5 Conclusion

In this thesis we evaluated if GARCH-type models can improve the foreign exchange VaR compared to Historical Simulation, and which part of the model (the variance dynamics, the asymmetry or the innovation distribution) is responsible for the improvement. We used a design of nine specifications on three euro pairs and a strict out-of-sample backtesting procedure over the period 2019–2026.

The answer to the research question is yes, but with some conditions. Modelling the conditional volatility improves the risk estimation: Historical Simulation gives more or less the correct number of violations, but it fails the Dynamic-Quantile test, so its violations stay predictable. However, the improvement is not the same for every modelling choice. The **innovation distribution is the most important factor**: Student- t and skewed Student- t are preferred in-sample by AIC/BIC for all the three pairs, and, more importantly, they give an adequate 99% VaR out

of sample, while the Gaussian models do not cover the deep tail. The **asymmetry adds little**: for EUR/USD the symmetric GARCH and GJR pass every test, while the EGARCH is rejected, and the estimated leverage parameters are very small; only for the jump-prone EUR/GBP an asymmetric model wins, and there because of its skewed heavy tail and not because of the leverage term. So the model that we recommend for these euro pairs is a simple **GARCH(1,1) or GJR-GARCH(1,1) with Student- t innovations**. This conclusion is close to Hansen and Lunde [7]: for exchange rates it is difficult to beat a well-specified GARCH(1,1), and the important thing is to model correctly the tail of the innovation distribution and not to make the variance equation more complicated.

Limitations and extensions. Our work has of course some limitations, and they suggest some natural extensions. First, we run the backtests one specification at a time, without a formal correction for multiple testing. A Model Confidence Set [8] or a Superior Predictive Ability test on a quantile-tick loss would make the “horse-race” more rigorous. Second, VaR does not say anything about how large the loss is when the threshold is exceeded. The 97.5% *Expected Shortfall* of the Basel III framework, or Extreme-Value and jump-diffusion models, would be the next step, especially for EUR/GBP, whose deep tail is dominated by a single event. Third, our Historical Simulation benchmark is on purpose very simple; a filtered Historical Simulation or an EWMA/RiskMetrics model would be a more fair parametric competitor. Fourth, the study uses only three euro pairs and one post-2015 sample; the result that the asymmetry is not important should hold for liquid FX but not for equities, where the leverage effect is real, and the generalization to more currencies or to longer horizons is still open. But none of these points changes the main message: for euro FX risk, a simple and robust volatility model together with a fat-tailed innovation distribution, validated out of sample, is enough and is better than more complicated alternatives.

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